

TIRE: Advancing Threat Intelligence Relation Extraction with a Novel Data-Centric Framework

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Abstract. Relation extraction (RE) plays a critical role in uncovering hidden connections and assisting security analysts in identifying complex patterns within cyber threat intelligence (CTI) data. Despite its importance, RE faces significant challenges, such as overlapping relations, complex sentence structures with long-range dependencies, and ambiguous relation types. Existing solutions primarily rely on model-centric approaches based on entity-marked and entity-tagging representations. However, these methods require modifying the original text and model architecture, increasing complexity. Furthermore, they fail to provide adequate contextual and semantic information, leading to suboptimal performance, particularly in joint extraction settings. This research introduces TIRE, a data-centric framework that addresses these challenges through an innovative multi-sequence representation (MSR) for the RE. By incorporating key features such as Entity Mask and Entity Type, TIRE enhances contextual and semantic understanding, enabling precise classification of relationships between entities. Unlike complex model-centric approaches, TIRE achieves state-of-the-art performance with simplified architectures. Extensive evaluations on the curated DNRTI-AUG-STIX2-JE dataset demonstrate TIRE’s superior performance in both pipeline and joint extraction settings, consistently achieving an F1 score of 99% in RE tasks while maintaining computational efficiency. TIRE’s innovative design bridges the gap between NER and RE tasks for constructing high-quality cybersecurity knowledge graphs (CKGs) and shows adaptability to domains like finance, healthcare, and biomedical fields where structured information extraction is critical. This work underscores the potential of data-centric designs to advance relation extraction and support real-world applications.

Keywords: Threat Intelligence Relation Extraction (TIRE) · Multi-Sequence Representation (MSR) · Cybersecurity Knowledge Graphs · Joint Extraction · Pipeline Extraction

1 Introduction

Cyber threat intelligence (CTI) analyzes threat data to generate actionable insights, structured into cybersecurity knowledge graphs (CKGs) or provenance graphs. CKGs integrate diverse data sources, automating threat analysis, attack

path visualization, and proactive defense strategies. Their growing adoption addresses skill shortages, enhances service quality, and improves security analysts' workflow and efficiency [5,6,8]. Key information retrieval methods for constructing CKGs include named entity recognition (NER) and relation extraction (RE), implemented via joint or pipeline approaches. Joint extraction (JE) integrates NER and RE into a single model, leveraging task interactions to enhance performance, while pipeline extraction (PE) processes them sequentially [8,6]. NER identifies threat entities (e.g., actors, malware, vulnerabilities, tools), while RE defines relationships between them (e.g., uses, targets, hasVulnerability). Entities (nodes) and relationships (edges) in CKGs are often based on standardized schemas like the unified cybersecurity ontology (UCO) 2.0 [4] and structured threat intelligence expression (STIX) [9].

In cybersecurity, RE is crucial for uncovering hidden connections, identifying attack patterns, and predicting future threats. However, accurate relationship extraction is hindered by challenges such as overlapping relations, long-range dependencies, and ambiguous relation types.

Overlapping entities and relations: RE assigns one relation per entity pair, yet a sentence may contain multiple entity pairs with overlapping entities in different relations [10,6,7]. The model must accurately distinguish these relationships by understanding sentence context and entity semantics.

Complex sentence structures and long-range dependencies: Entities in a relation may be separated by many tokens, requiring hybrid and complex models [11,6] to effectively capture dependencies while maintaining sentence coherence.

Subtle and ambiguous relation types: Security reports use domain-specific terms that lack clear natural language meanings. For example, "hasAttackTime" links a threat actor to a time, while "hasVulnerability" connects software to a vulnerability. Though meaningful to analysts, such terms pose challenges for models in distinguishing relationship types accurately [11].

Two sentence representation methods are widely used in threat intelligence for RE: entity-marked and entity-tagging representations. The entity-marked representation (Figure 1) generates separate sequences for each entity pair, using explicit markers (`<e1>`, `</e1>`, `<e2>`, `</e2>`) to highlight entities [8,13,12]. While effective in pipeline extraction, it requires modifying the text and model architecture, increasing complexity and training parameters. This makes it less suitable for JE, where NER and RE are learned simultaneously. The entity-tagging representation (Figure 2), commonly used in JE [7,6,10], employs tagging schemes like BIO (B: beginning, I: inside, O: outside) to label entity pairs. In a "uses" relation, only "APT29" and "Mimikatz" are tagged, while other tokens receive "O." This relation-specific labeling can disrupt the NER task, as tokens lose their original labels during training.

Although both representations effectively manage overlapping relations [6,7], they lack contextual and semantic depth, limiting classification performance. This drawback is especially pronounced in JE, where relation classification F1 scores remain low despite advanced hybrid models.

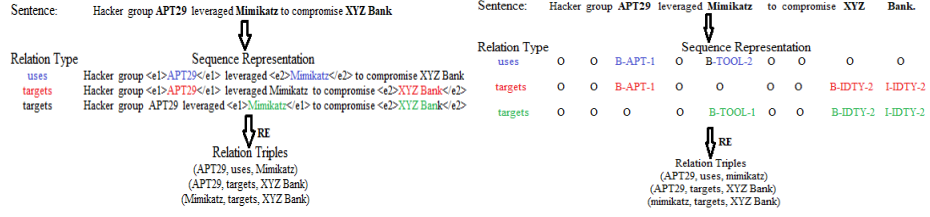


Fig. 1: Entity-marked Representation. Fig. 2: Entity-tagging Representation.

As cybersecurity datasets grow in complexity and automation becomes essential, effective RE methods that simplify model architectures are needed. This paper presents TIRE, a novel framework for threat intelligence relation extraction. Taking a data-centric approach, TIRE introduces an innovative sequence tagging representation to handle overlapping entities and relations in both PE and JE settings. By integrating data-driven features like entity masks and entity types, TIRE enhances contextual understanding, mitigates language ambiguities, and reduces relation type confusion. These enhancements allow TIRE to achieve high relation classification accuracy while maintaining a streamlined model design. The key contributions of this work can be summarized as follows:

- Introduces TIRE, a novel data-centric framework for relation extraction in CTI. TIRE employs multi-sequence representation (MSR) to tackle key challenges such as overlapping relations, complex sentence structures, and ambiguous relation types. MSR incorporates Entity Mask, which tracks entity pair positions to enhance focus, and Entity Type, which provides essential contextual and semantic cues for accurate relation classification. TIRE significantly improves RE accuracy, reduces reliance on complex architectures, and offers a practical and efficient cybersecurity solution.
- Develops DNRTI-AUG-STIX2-JE, a curated dataset for joint entity and relation extraction, addressing the scarcity of publicly available annotated datasets for this task. Derived from DNRTI-AUG-STIX2 [29], initially released for NER-only tasks, DNRTI-AUG-STIX2-JE will be made publicly available on GitHub³ to support reproducibility and future research in threat intelligence.
- Conducts comprehensive evaluations using DNRTI-AUG-STIX2-JE, a domain-specific dataset aligned with the STIX-2.1 standard, validating TIRE’s performance in both pipeline and joint settings. TIRE demonstrates high computational efficiency and superior performance with simplified architectures, proving its suitability for real-world applications.
- Assesses SecureBERT⁺ [3], which reportedly improves masked language modeling (MLM) performance by 9% over SecureBERT [2]. However, our results indicate that this improvement does not significantly enhance NER and RE tasks.

³ <https://github.com/imouiche/TIRE>

The remainder of this paper is organized as follows: Section 2 reviews existing relation extraction approaches and highlights key gaps in the current research. Section 3 details the methodology for constructing the TIRE model and its evaluation and comparison against state-of-the-art approaches. Finally, Section 4 summarizes the findings and discusses potential directions for future research.

2 Related Works

Relation extraction (RE) identifies relationships between entity pairs, either in a binary setting (determining if a relation exists) or an n-ary setting (common in cybersecurity), where a relation is classified from a predefined set. Most RE studies follow a model-centric approach, relying on models to extract features for classification. This paper, however, adopts a data-centric approach, leveraging a tailored representation to expose key features, improving model decision-making and RE accuracy. This section reviews RE studies in both pipeline and joint settings, highlighting the techniques and methods employed for the RE tasks.

2.1 Relation Extraction in the Pipeline Approach

In pipeline extraction (PE), also known as mention-level extraction, RE predicts relationships between pre-annotated entities within a sentence [19]. Satvat et al.[15] developed EXTRACTOR to identify system-level indicators of compromise (IoCs) (e.g., processes, files, registries) using POS tagging, dependency parsing (DP), and semantic role labeling to construct provenance graphs. Gao et al.[16] applied a DP tree algorithm in their THREATRAPTOR framework to extract relationships among log-based IoC entities from system audit logs. Zhao et al. [18] leveraged heterogeneous information networks (HINs) and syntactic dependency parsing to define templates for extracting nine interdependent relationships among IoC entities from CTI reports. While effective, these rule-based approaches rely heavily on syntactic structures, lacking contextual and semantic understanding, which limits their ability to handle overlapping entities and multiple relationships. Gasmi et al.[17] combined LSTM networks, shortest dependency paths, POS tags, and word dictionaries to establish relationships in vulnerability descriptions. Jo et al.[20] introduced Vulcan, which employs BERT with fully connected layers to classify threat relationships between malware entities. Similarly, Mouiche and Saad [8] extracted 15 relation types from unstructured APT reports using GloVe embeddings, BiLSTM, and an attention mechanism. Mouiche et al.[29] introduced a hybrid model integrating SecureBERT embeddings to enhance relation classification accuracy. Studies such as [17,20,8,29] employed entity-marked representation (Figure 1), modifying text and model architecture with entity markers. While effective for handling overlapping relations, this approach increases model complexity. Additionally, models in [17] and [8] rely on static embeddings (e.g., GloVe, dictionary-based embeddings), where word representations remain unchanged across contexts. This leads to

ambiguity and reduced RE performance, as context-specific meanings are crucial for accurate extraction. These limitations highlight the need for dynamic approaches like TIRE, which leverages contextualized embeddings and a novel entity representation. TIRE integrates semantic entity information, improving inter-entity understanding and significantly enhancing RE accuracy. Using verb similarity functions, Marchiori et al. [21] combined rule-based and deep learning methods to extract relationships without predefined relation types. However, their method struggles with overlapping entities and false positives, particularly when entities are separated by several tokens, as identical verbs can imply different relations based on context. Li et al. [22] used a BERT attention-based model with POS tags, entity types, and entity pair distances for relation extraction. However, their document-level RE model lacks robust representation for overlapping relations, leading to a low F1 score (60%), which impacts knowledge graph quality. Sarhan and Spruit [23] developed Open-CyKG, a cybersecurity knowledge graph framework using open information extraction (OIE) with static embeddings and sentence predicates. However, this approach lacks contextual nuance and struggles with implicit relationships, reducing its effectiveness in accurately linking cyber entities. Wang et al. [25] proposed KnowCTI, a pipeline for extracting APT entity relationships. It relies on preconstructed ontologies and fuzzy knowledge matching, limiting its ability to capture contextual and semantic nuances. This constraint leads to challenges in relation classification and a moderate F1 score (81%) for RE.

Piplai et al. [24] developed RelEx, a neural network-based RE model using Word2Vec embeddings. It employs an input layer, three hidden layers, and a softmax output layer to classify entity pairs into six predefined relations. However, static embeddings limit RelEx’s ability to capture contextual variations, reducing its accuracy in dynamic scenarios. TIRE overcomes these limitations by integrating entity masks and entity types, embedding expert knowledge into the model. This approach enhances semantic focus and contextual understanding during training, significantly improving relation classification accuracy.

2.2 Relation Extraction in Joint Settings

In joint extraction (JE), NER and RE are integrated into a single model, allowing simultaneous learning to improve classification accuracy. Unlike pipeline extraction (PE), which suffers from error propagation, JE enhances efficiency and accuracy in extracting entity-relation triples from unstructured CTI reports, making it a preferred approach for cybersecurity knowledge graph construction. Guo et al. [7] proposed CyberRel, a joint model using an entity-tagging representation with BERT, BiGRU, and attention for feature extraction. Similarly, Zuo et al. [10] and Ahmed et al. [6] adopted sequence tagging and external relation matching to improve relation classification accuracy.

Despite using entity-tagging representation to handle overlapping relations, these studies report low F1 scores for RE (72% for [10] and 79% for [6]), even with advanced NLP models. The tagging approach (Figure 2) lacks contextual depth,

limiting relation prediction accuracy. To compensate, Zuo et al.[10] and Ahmed et al.[6] introduced rule-based techniques, questioning the true benefits of joint learning in their models.

You et al. [26] introduced TiGNet, a joint extraction model using token-pair matrices for token span representation. While effective for overlapping relations, it focuses on local interactions, potentially missing broader contextual cues. Designed for Chinese text, TiGNet achieves a 71% F1 score but introduces computational complexity due to its reliance on multiple matrix interactions. Liu et al. [27] introduced CTI-JE, integrating SecBERT-BiLSTM embeddings with a multi-layer perceptron to enhance relation classification. The model leverages distance embeddings and contextual features to capture relationships. However, many cybersecurity relations are implicit or require background knowledge beyond the sentence context, limiting the table-filling method’s effectiveness, especially for nuanced relation types. Wang et al. [11] proposed the Enhanced Representation and Binary Tagging Framework (ERBTF), which integrates relation and word embeddings to enhance classification. It employs two binary taggers to identify subject-object entity pairs. However, ERBTF supports only single overlapping entities, limiting its performance. Additionally, it requires manual replacement of complex relation types with simpler terms, a subjective and error-prone process that reduces generalization and real-world applicability. Unlike previous studies, TIRE employs a data-centric approach to dynamically extract sentence-level contextual and semantic features, ensuring robust entity-relation interactions and significantly enhancing RE accuracy.

3 TIRE Methodology

Unlike existing model-centric approaches, TIRE takes a data-driven approach to tackle key RE challenges, such as overlapping relations, complex sentence structures, and ambiguous relation types. To ensure a clear and logical flow, the methodology first introduces the dataset, followed by feature engineering, sequence tagging representation, model architecture, and implementation.

3.1 Dataset

The DNRTI-AUG-STIX2 dataset, recently released by Mouiche et al. [29], extends DNRTI-STIX2 [28] and was selected for its compliance with the STIX-2.1 standard and its rich diversity of entity categories (19 in total), including APT groups, malware, tools, vulnerabilities, and attack techniques. It contains 43,258 labeled entities across 7,947 sentences and supports 16 predefined relation types, such as targets, uses, hasVulnerability, and affiliatedWith. DNRTI-AUG-STIX2 provides a realistic simulation of complex cyber threat scenarios, making it ideal for training models to generalize effectively in relation extraction tasks. The dataset consists of APT reports sourced from MITRE ATT&CK and other threat intelligence repositories [29,28], ensuring its real-world applicability in cybersecurity.

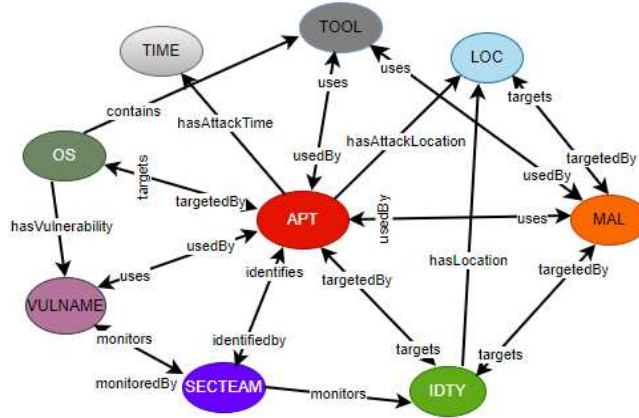


Fig. 3: A sample ontology schema containing entities (nodes) and 12 relation categories (edges).

The DNRTI-AUG-STIX2 dataset was initially released for NER-only tasks, with entities annotated using the BIO tagging format. In this work, we extend the dataset by annotating entity relations, assigning relation types based on a predefined domain-specific ontology. Figure 3 illustrates a sample schema, displaying 9 entities (nodes) out of 19 and 12 relations out of the 16 categories for clarity. For a comprehensive ontology description, we adopt definitions from the Unified Cybersecurity Ontology (UCO)[4], a framework also utilized in prior studies[8,24,29]. This ontology specifies the Domain and Range for each of the 16 predefined relation types, ensuring consistency in entity-relation mapping.

- affiliatedWith: Domain:APT
Range:APT
- associatedWith: Domain:HASH or VULNAME or VULID
Range: EMAIL or ACT or ENCR or DOM or URL or TOOL or OS or PROT
- contains: Domain: FILE or EMAIL
Range:MAL or IP or URL
- hasAttackLocation: Domain:APT or MAL or ACT
Range:LOC
- hasAttackTime: Domain: APT or MAL or ACT
Range:TIME
- hasLocation: Domain: IDTY or SECTEAM
Range:LOC
- hasVulnerability: Domain:IDTY or OS or URL or DOM or PROT or FILE
Range:VULID or VULNAME
- identifies: Domain:SECTEAM
Range: APT or MAL or VULNAME or VULID or ACT
- identifiedBy: Domain:APT or MAL or VULNAME or VULID or ACT
Range:SECTEAM

- monitors: Domain:SECTEAM
Range:IDTY or DOM or FILE
- monitoredBy: Domain:IDTY or LOC or FILE
Range:SECTEAM
- targets: Domain:APT or MAL or ACT
Range:IDTY or DOM or VULNAME or VULID or OS
- targetedBy: Domain:IDTY or DOM or VULNAME or VULID or OS or LOC
Range:APT or MAL or ACT
- uses: Domain:APT or MAL or ACT
Range:EMAIL or IP or URL or FILE or TOOL or HASH or ENCR or MAL or ACT
- usedBy: Domain: EMAIL or IP or URL or FILE or TOOL or HASH or ENCR or MAL or ACT
Range: APT or MAL or ACT

The noRelation type represents entity pairs with no predefined relationship (e.g., TIME and HASH). The resulting dataset, DNRTI-AUG-STIX2-JE, will be publicly available via our GitHub repository, addressing the scarcity of open-access datasets for joint entity and relation extraction in cybersecurity. Table 1 presents the distribution of entities and relations, with 42,950 annotated relations, making DNRTI-AUG-STIX2-JE the benchmark dataset for evaluating the TIRE framework.

3.2 Feature Engineering

This subsection identifies key features that encode expert knowledge to capture entity-relation interactions effectively. These features provide contextual, semantic, and positional information, enabling the model to make expert-informed relation predictions. To derive these features, we analyze an annotated example sentence (Figure 4), which includes three relation triples: (APT29, uses, Mimikatz), (APT29, targets, XYZ Bank), and (Mimikatz, targets, XYZ Bank). Consider

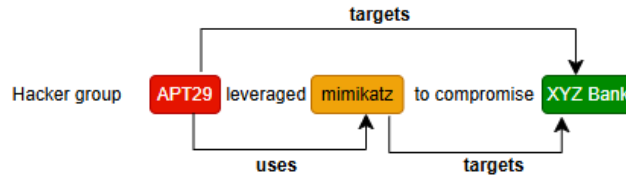


Fig. 4: Graphical representation of a sample annotated sentence containing an overlapping relation to type "targets".

the first relation triple (APT29, uses, Mimikatz). The key challenge is determining what guides a security analyst to classify this relation as "uses" rather than

Table 1: DNRTI-AUG-STIX2-JE dataset with the number of entity and relation types.

Entity Type	Count	Relation Type	Count
ACT	9070	affiliatedWith	1201
APT	5906	associatedWith	3067
DOM	435	contains	1043
EMAIL	66	hasAttackLocation	2553
ENCR	242	hasAttackTime	1444
FILE	2458	hasLocation	1652
HASH	483	hasVulnerability	1262
IDTY	5845	identifies	1797
IP	229	identifiedBy	978
LOC	3615	noRelation	8510
MAL	3924	monitors	1050
OS	600	monitoredBy	837
PROT	519	targets	4933
SECTEAM	2026	targetedBy	2630
TIME	3039	uses	7451
TOOL	3898	usedBy	2542
URL	100		
VULID	803		
VULNAME	1312		
# of entity types	19	# of relation types	16
# of labeled entities	43,258	# of labeled relations	42,950

alternatives like "affiliatedWith" or "associatedWith". The crucial factor is the context and semantic meaning of APT29 and Mimikatz in cybersecurity. APT29 is a hacker organization, while Mimikatz is a tool used for cyberattacks. This distinction is captured through their Entity Types (APT for hacker organization and TOOL for a tool), which directly influences relation prediction. Without this context, a model relying solely on literal meanings could misinterpret APT29 as an apartment or Mimikatz as an arbitrary term. Since Mimikatz lacks a general vocabulary definition outside cybersecurity, such ambiguity introduces feature confusion, making it harder for generic language models to classify relationships accurately. The same reasoning applies to the other relation triples in the sentence, highlighting the crucial role of the Entity Type feature. By encoding essential contextual and semantic information, Entity Type ensures the model correctly interprets entity relationships within the cybersecurity domain. This prevents the model from being misled by generic or ambiguous interpretations, allowing it to make accurate, security-specific relation classifications.

Another key feature introduced in this paper is the Entity Mask, which tracks the positions of entity pairs involved in a given relation. Represented as a binary vector, the Entity Mask assigns "1" to tokens within the entity pair and "0" to all other tokens. This feature provides precise positional information, allowing the model to focus on relevant parts of the sequence during relation classification.

As illustrated in Figure 5, the Entity Mask highlights entity tokens for each generated sequence, enhancing the model’s ability to localize and process entity interactions. This improves relation extraction accuracy, particularly in complex sentence structures where entity dependencies span multiple tokens.

3.3 Multi-Sequence Representation

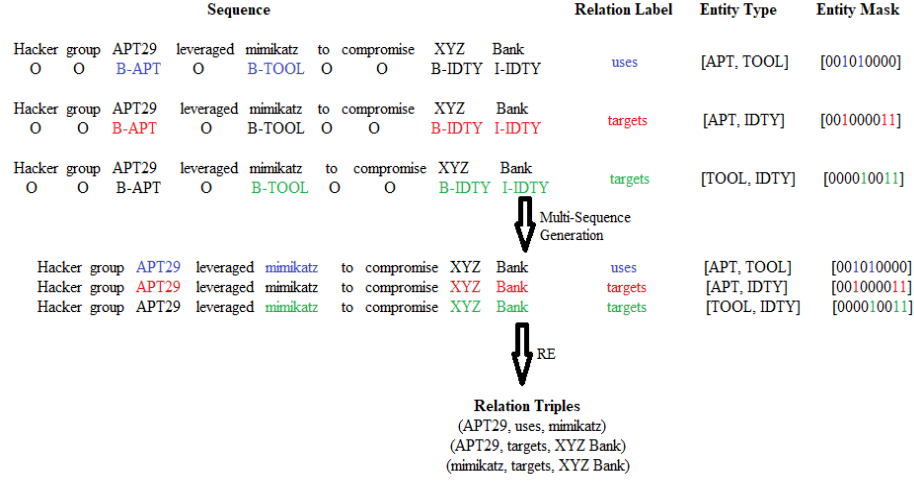


Fig. 5: Multi-sequence generation and representation for the relation extraction.

Figure 5 illustrates the Multi-Sequence Representation (MSR) for relation extraction, applicable to both joint and pipeline settings. Unlike Entity-Marked Representation (EMR) (Figure 1) used in PE approaches [8,17], MSR eliminates the need for entity markers, avoiding modifications to the text or model architecture, which can increase complexity. Instead, MSR leverages the Entity Mask to track entity positions while preserving the original input format. Similarly, compared to Entity-Tagging Representation (ETR) (Figure 2) used in JE approaches [6,7,10], MSR maintains the original token labels within the sentence, which is essential for NER consistency. Additionally, MSR uses the Entity Mask to highlight entity positions, enabling precise tracking of relational context without disrupting entity recognition. Furthermore, MSR integrates the Entity Type feature, embedding crucial contextual and semantic information to improve relation classification. This structured representation supports both JE and PE techniques, reducing dependency on model embedding layers for feature extraction. By streamlining input representation, MSR enhances RE accuracy while maintaining simplicity and compatibility across various model architectures. MSR effectively addresses key RE challenges by:

Overlapping Relations: Like EMR and ETR, MSR generates separate sequences for each entity pair, effectively managing overlapping relationships.

Complex Sentence Structures or Long-Range Dependencies: MSR leverages the Entity Mask to highlight entity positions, enabling the model to maintain focus on relevant entities even in complex sentence structures. Unlike EMR, MSR preserves the original text and architecture, reducing complexity and improving performance. Additionally, unlike ETR, MSR retains original token and entity labels, making it ideal for JE techniques.

Ambiguity and Subtle Relation Types: Non-vocabulary terms (e.g., hasAttackTime) challenge model-centric techniques that rely on semantic meaning for relation classification [11]. Overlapping relations require sentence duplication for each relation type, generating multiple identical sequences with different relation labels. This issue worsens with complex relations (e.g., hasVulnerability), which lack clear semantic cues, making classification difficult. By integrating Entity Type, MSR enriches contextual understanding, allowing the model to differentiate relationships more effectively. This mitigates ambiguity and enhances classification accuracy for subtle and complex relations.

3.4 TIRE Model Architecture

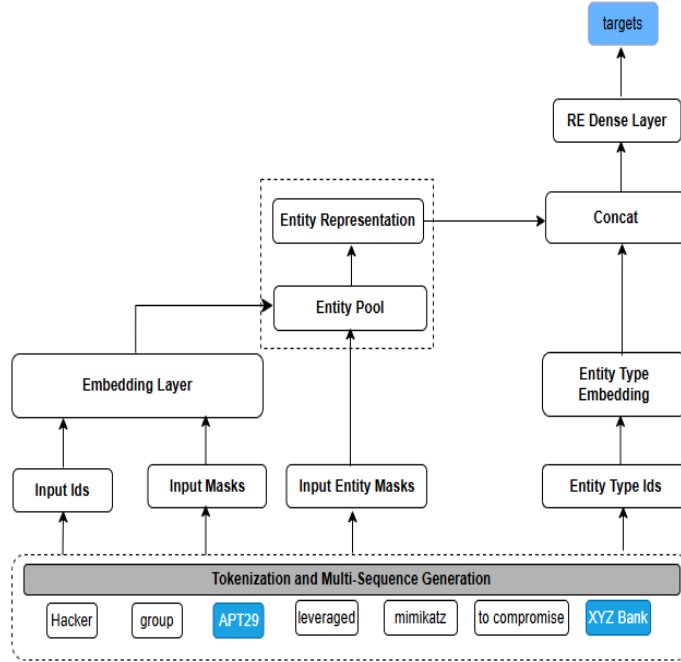


Fig. 6: TIRE model’s architecture for relation extraction.

The TIRE model architecture, illustrated in Figure 6, is optimized for relation extraction (RE). It leverages contextual embeddings from either general large

language models (LLMs) (e.g., BERT, RoBERTa) or domain-specific embeddings (e.g., SecureBERT, SecureBERT⁺) to enhance token representation. Additionally, entity-specific features, such as entity pooling and entity type embeddings, are integrated to improve relation classification. The architecture comprises six interconnected layers, mathematically formulated as follows:

1. Tokenization and Multi-Sequence Representation (MSR):

This layer is essential for dynamically and automatically generating the MSR for each input sentence. It produces the following inputs for each generated sequence:

- $\mathbf{X} = [x_1, x_2, \dots, x_T]$: The input sentence is tokenized into T tokens.
- $\mathbf{I}_{\text{mask}} \in [0, 1]^T$: Represents the input mask of the sequence, where each position i has a value of 1 if x_i is a valid token, and 0 if it is padding.
- $\mathbf{I}_{\text{entity_mask}} \in [0, 1]^T$: Highlights or tracks the positions of both entities within the sentence that are involved in a given relation type.
- $\mathbf{I}_{\text{entity_type}} \in \mathbb{Z}^n$: Represents the label IDs for different entity types, where $n = 19$ corresponds to the total number of entity categories. Each entity type is mapped to a unique integer in this set.

2. Embedding Layer:

Applies pretrained embeddings to generate token representations:

$$\mathbf{H} = [h_1, h_2, \dots, h_T], \quad h_i \in \mathbb{R}^d$$

where d is the embedding dimension.

3. Entity Pooling Layer:

Extracts entity representations by pooling contextual embeddings:

$$h_{\text{entity1}} = \sum_{i \in \text{Entity1}} h_i \cdot \mathbf{I}_{\text{entity_mask}, i} \quad (1)$$

$$h_{\text{entity2}} = \sum_{i \in \text{Entity2}} h_i \cdot \mathbf{I}_{\text{entity_mask}, i} \quad (2)$$

yielding:

$$\mathbf{H}_{\text{entity}} = [h_{\text{entity1}}, h_{\text{entity2}}]$$

4. Entity Type Embedding Layer:

Encodes entity types as static embeddings:

$$e_{\text{type1}}, e_{\text{type2}} = \text{Embedding}(\mathbf{I}_{\text{entity_type}}) \quad (3)$$

resulting in:

$$\mathbf{E}_{\text{type}} = [e_{\text{type1}}, e_{\text{type2}}]$$

5. Feature Concatenation:

Combines entity and type embeddings:

$$\mathbf{H}_{\text{concat}} = \text{Concat}(h_{\text{entity1}}, e_{\text{type1}}, h_{\text{entity2}}, e_{\text{type2}}) \quad (4)$$

6. Relation Classification Layer:

Performs classification via a softmax layer:

$$\hat{y} = \text{Softmax}(W_{\text{RE}} \cdot \mathbf{H}_{\text{concat}} + b_{\text{RE}}) \quad (5)$$

where:

$$\hat{y} \in \mathbb{R}^{\text{num_re_labels}}$$

represents relation probabilities.

Training Objective: Cross-Entropy Loss

We applied the cross-entropy loss function (Eq. (6)) to quantify the difference between predicted and true relation labels in the classification task.

$$\mathcal{L}_{\text{RE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^C y_i^{(j)} \log \hat{y}_i^{(j)} \quad (6)$$

where N is the batch size, and C is the number of relation classes.

The probability of a relation class is computed as:

$$\hat{y}_i^{(j)} = \frac{\exp(z_i^{(j)})}{\sum_{k=1}^C \exp(z_i^{(k)})} \quad (7)$$

where $z_i^{(j)}$ is the raw logit for class j for sample i .

3.5 Implementation

Dataset Format: The DNRTI-AUG-STIX2-JE dataset, used to evaluate TIRE, is stored in JSON format (see Figure 7 showing a sample data entry) with the following structure:

- **text:** A sentence that describes a specific attack scenario.
- **entities:** A list of annotated entities with their start and end indices.
- **relations:** A list of identified relationships for each valid entity pair in the "entities" list. The indices refer to the positions of the head and tail entities within the "entities" list.
- **entity_labels:** BIO-tagged labels for each token, denoting entity types or "O" for non-entities.

Notably, the dataset does not explicitly store Entity Mask and Entity Type features. Instead, TIRE includes a module that automatically transforms input samples into the MSR representation required for model training.

Baselines: To assess TIRE’s performance, we implemented state-of-the-art RE approaches from scratch, replicating effective methods from existing studies. While prior works rely on complex model-centric techniques for feature extraction, TIRE adopts a data-centric approach with multi-sequence representation. Our results show that TIRE achieves comparable RE performance using a simpler model, demonstrating that high classification accuracy can be attained without overly complex architectures.

```
[...
{'text': 'Hacker group APT29 leveraged mimikatz to compromise XYZ Bank',
 'entities': [[3, 4, 'APT29', 'APT'], [5, 6, 'mimikatz', 'TOOL'],
              [8, 10, 'XYZ Bank', 'IDTY']],
 'relations': [['uses', 0, 1], ['targets', 0, 2], ['targets', 1, 2],
 'entity_labels': ['O', 'O', 'B-APT', 'O', 'B-TOOL', 'O', 'O', 'B-IDTY', 'I-IDTY']
 },
 {...},
 ...
 ]
```

Fig. 7: A sample data entry in json format.

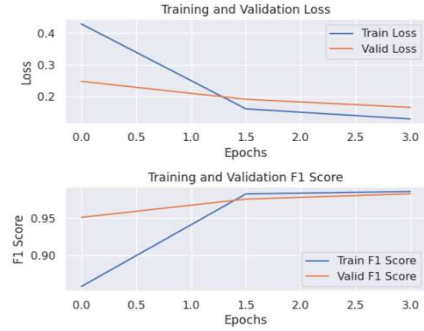


Fig. 8: TIRE model’s training and validation losses and F1 Scores using SecureBERT⁺.

Table 2: RE Models’ parameter settings.

parameters	BERT Variants	BiLSTM	CNN
batch size	16	16	16
dropout	0.3	0.3	0.3
epsilon	1e-8	1e-8	1e-8
laerning_rate	5e-5	5e-5	5e-5
hidden_size	256 × 2	100 × 2	-
embedding_size	768	300	300
# of epochs	3	50	100

Training and Evaluation: The DNRTI-AUG-STIX2-JE dataset was split into training (70%), validation (15%), and test (15%) sets for training and evaluation. Table 2 details the base model parameters. All experiments were conducted on an NVIDIA Tesla T4 with CUDA 12.1, ensuring efficient and consistent execution. Table 3 highlights a detailed comparison of the TIRE framework with various state-of-the-art model-centric approaches based on EMR. Several critical insights:

- **Superior F1 Score:** TIRE consistently achieves the highest F1 scores across all LLM embeddings, with SecureBERT⁺ and SecureBERT reaching 99% (0.989), demonstrating its effectiveness in leveraging MSR and entity-centric features to enhance relation classification accuracy. The best-performing model-centric baselines, SecureBERT-BiGRU-FC-CRF and SecureBERT⁺-BiGRU-FC-CRF, achieve 96%, yet TIRE surpasses them, underscoring its efficiency and performance advantage. Even with a basic BERT model, TIRE outperforms these baselines by integrating MSR, which encodes positional and semantic context through Entity Type and Entity Mask features. This structured representation enables a better understanding of entity-relation interactions, reducing reliance on complex architectures. Additionally, TIRE enhances contextual awareness by explicitly associating entity types with

Table 3: Performance comparison of the TIRE model in the PE settings for RE.

Model Centric based on EMR				
Models	Precision	Recall	F1	Elapsed Time
Glove-CNN-TDD+Att [8]	0.845	0.843	0.844	1h 40m 11s
BERT-BiLSTM-CRF [20]	0.903	0.895	0.895	1h 30m 20s
BERT-BiGRU-CRF [7,10]	0.914	0.905	0.897	1h 28m 17s
Glove-BiLSTM-TDD+Att [8]	0.908	0.912	0.913	1h 40m 09s
BERT-DCNN+Att [11]	0.923	0.930	0.926	1h 32m 09s
RoBERTa-BiGRU-FC-CRF [6]	0.942	0.953	0.945	1h 34m 40s
SecureBERT-BiGRU-FC-CRF [29]	0.955	0.957	0.956	1h 31m 07s
SecureBERT ⁺	0.933	0.938	0.941	1h 27m 23s
SecureBERT ⁺ -BiGRU-FC-CRF	0.954	0.958	0.958	1h 33m 43s
TIRE's Framework				
LLM Embedding	Precision	Recall	F1	Elapsed Time
BERT	0.976	0.969	0.973	42m 33s
RoBERTa	0.976	0.974	0.976	51m 01s
SecureBERT	0.986	0.991	0.989	50m 34s
SecureBERT ⁺	0.992	0.989	0.989	51m 03s

their cybersecurity-specific categories. Unlike conventional models that may misinterpret domain-specific terms like "Mimikatz", TIRE ensures accurate classification by reinforcing structured contextual cues, improving prediction accuracy even in complex cybersecurity.

- **Improved Efficiency:** The TIRE framework demonstrates superior computational efficiency compared to baseline models. SecureBERT⁺ within TIRE completes the task in 51 minutes and 3 seconds, while the BERT base model finishes in 42 minutes and 33 seconds, outperforming all baselines and reducing processing time by over 30 minutes. This suggests that model-centric approaches require more time for sentence feature extraction, whereas TIRE efficiently encodes entity-relation interactions through MSR, minimizing computational overhead. This efficiency highlights TIRE's scalability and suitability for real-world applications, particularly in large-scale cybersecurity datasets where fast processing is essential.
- **Simplified Model Architecture:** Model-centric approaches rely on complex architectures like BiLSTM, BiGRU, CRF, and attention mechanisms, increasing model complexity and training time. In contrast, TIRE adopts a data-centric approach that simplifies the architecture by leveraging features like the Entity Mask and Entity Type, which encode contextual and semantic information without additional layers for feature extraction. This reduces computational overhead while maintaining high performance.
- **SecureBERT⁺ vs. SecureBERT:** While SecureBERT⁺[3] has been reported to achieve a 9% improvement over SecureBERT in masked language modeling (MLM) tasks, our findings show no significant performance gains for NER and RE. This suggests that improvements in MLM do not nec-

essarily enhance structured tasks like entity and relation extraction, which require deeper contextual understanding beyond masked word prediction.

Figure 8 presents the training and validation loss and F1 scores over three epochs for TIRE with SecureBERT⁺. The training loss decreases rapidly in the first epoch and stabilizes by the third, indicating effective optimization and convergence. Similarly, the validation loss follows a steady decline with minimal gap between the curves, suggesting strong generalization and low risk of overfitting. The F1 score curves show a sharp increase after the first epoch, with validation F1 closely tracking the training F1. By the third epoch, both scores nearly align, demonstrating the model’s robustness and ability to generalize well to unseen data. This paper also evaluates TIRE in a joint extraction setting, as shown

Table 4: Performance comparison of the TIRE model in the JE settings.

LLM Embedding	NER			RE			Elapsed Time
	Precision	Recall	F1	Precision	Recall	F1	
BERT	0.846	0.852	0.843	0.967	0.974	0.968	1h 40m 05s
RoBERTa	0.877	0.886	0.892	0.979	0.978	0.976	1h 53m 15s
SecureBERT	0.910	0.915	0.907	0.988	0.991	0.986	1h 53m 40s
SecureBERT ⁺	0.905	0.923	0.917	0.992	0.987	0.991	1h 55m 23s

in Table 4, where it achieves state-of-the-art F1 scores for both NER and RE tasks, particularly with domain-specific embeddings like SecureBERT⁺ and SecureBERT. TIRE maintains computational efficiency while delivering consistent RE accuracy across both pipeline and joint extraction approaches (see Tables 4 and 3), reinforcing its practicality for cybersecurity applications. While stacking BiGRU, BiLSTM, or CRF layers could further enhance NER performance, this study prioritizes improving RE accuracy through a novel approach, leaving such extensions for future research. TIRE enables seamless interaction between NER and RE in joint settings, significantly enhancing performance in both tasks.

Future research can explore other datasets, such as STUCCO[1], which contains reports from vulnerability databases. Unlike APT reports, STUCCO is considered less complex, as machine learning models can easily capture patterns within vulnerability descriptions[8]. This difference in complexity could significantly impact NER performance, providing further insights into the adaptability of extraction models across different cybersecurity contexts.

3.6 Theoretical Implications and Limitations

The core of this data-driven RE approach lies in its key features: Entity Mask and Entity Type, which encode contextual, semantic, and positional information to enhance relation classification accuracy. The multi-sequence representation (MSR) serves as the mechanism that delivers this information, improving prediction accuracy. These features can be derived from any NER dataset, regardless of tagging schemes or labels, making the approach highly adaptable. Beyond cybersecurity, it applies to other domains, such as news datasets with

entity types like person, geoLocation, and date, or healthcare and biomedical datasets with entities like covid-19, fever, and symptoms. This demonstrates its versatility across diverse fields.

The main limitation is the reliance on expert knowledge for feature design and the complexity of data transformation. However, TIRE underscores the significance of these features in relation classification, offering valuable insights for cybersecurity and other domains requiring structured information extraction.

This research introduces a novel approach that mimics human reasoning to accurately identify hidden relationships between cyber entities, a crucial step in building high-quality CKGs. These CKGs enhance SOC efficiency by enabling automated threat analysis, proactive defense strategies, and improved incident response.

4 Conclusion

This research introduced TIRE, a data-centric framework for threat intelligence relation extraction that employs multi-sequence representation (MSR) and key features such as Entity Mask and Entity Type to improve contextual and semantic understanding. Unlike model-centric methods, TIRE achieves state-of-the-art performance in both pipeline and joint extraction settings without relying on complex architectures. Leveraging domain-specific embeddings like SecureBERT and SecureBERT⁺, TIRE demonstrated exceptional accuracy, scalability, and computational efficiency, surpassing existing baselines. Its robust design effectively handles overlapping relations, complex sentence structures, and ambiguous relationship types, making it a reliable solution for constructing high-quality cybersecurity knowledge graphs and supporting real-world cybersecurity applications. Future work will aim to extend TIRE’s capabilities by integrating advanced sequence modeling techniques, such as BiGRU and CRF, to enhance NER performance in joint settings. Additionally, efforts will explore TIRE’s adaptability to cross-domain applications, including healthcare and bioinformatics. Another promising direction involves applying TIRE’s approach to event and application logs, where features like temporal context could play a critical role. TIRE’s potential to bridge the gap between theoretical advancements and operational cybersecurity needs highlights its contribution to advancing the field of threat intelligence.

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References

1. Bridges, R. A., Jones, C. L., Iannacone, M. D., Goodall, J. R.: Automatic Labeling for Entity Extraction in Cyber Security. The Third ASE International Conference on Cyber Security 2014, (2014).

2. Aghaei, E., Niu, X., Shadid, W., Al-Shaer, E.: SecureBERT: A Domain-Specific Language Model for Cybersecurity. *Security and Privacy in Communication Networks*, vol. 462, (2023) <https://doi.org/10.1109/TrustCom50675.2020.00083>
3. Hugging Face, https://huggingface.co/ehsanaghaei/SecureBERT_Plu, last accessed 2025/02/25
4. Syed, Z., Padia, A., Finin, T., Mathews, L., Joshi, A.: UCO: A Unified Cybersecurity Ontology. *Proceedings of the AAAI Workshop on Artificial Intelligence for Cyber Security*, pp. 195–202, AAAI Press, (2016)
5. Arikkat, D.R., Vinod, P., K. A., R.R., Nicolazzo, S., Nocera, A., Conti, M. (2024). Relation Extraction Techniques in Cyber Threat Intelligence. In: Rapp, A., Di Caro, L., Meziane, F., Sugumaran, V. (eds) *Natural Language Processing and Information Systems. NLDB 2024. Lecture Notes in Computer Science*, vol 14762. Springer, Cham. https://doi.org/10.1007/978-3-031-70239-6_24
6. Ahmed, K., Khurshid, S., K., Hina, S.: CyberEntRel: Joint extraction of cyber entities and relations using deep learning. *Computers & Security*, vol. 136, (2024), <https://doi.org/10.1016/j.cose.2023.103579>
7. Guo, Y., Liu, Z., Huang, C., Liu, J., Jing, W., Wang, Z., Wang, Y.: CyberRel: Joint Entity and Relation Extraction for Cybersecurity Concepts. *Information and Communications Security: 23rd International Conference, ICICS 2021*, pp. 447–463, Springer, Chongqing, China (2021), https://doi.org/10.1007/978-3-030-86890-1_25
8. Mouiche, I., Saad, S.: Entity and relation extractions for threat intelligence knowledge graphs. *Computers & Security*, vol. 148, (2025), <https://doi.org/10.1016/j.cose.2024.104120>
9. OASIS OPEN, <https://docs.oasis-open.org/cti/stix/v2.1/cs02/stix-v2.1-cs02.html>, last accessed 2025/02/10
10. Zuo, J., Gao, Y., Li, X., Yuan, J.: An End-to-end Entity and Relation Joint Extraction Model for Cyber Threat Intelligence. 2022 the 7th International Conference on Big Data Analytics (ICBDA), pp. 204–209. IEEE, Guangzhou, China (2022), <https://doi.org/10.1109/ICBDA55095.2022.9760342>
11. Wang, X., Liu, Z., Liu, J.: Information extraction of cybersecurity concepts: an lstm approach. *Computers & Security*, vol. 144, (2024)
12. Zhou, P., Shi, W., Tian, J., Qi, Z., Li, B., Hao, H., Xu, B.: Attention-Based Bidirectional Long Short-Term Memory Networks for Relation Classification. *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics, Association for Computational Linguistics*, pp. 207–212, Berlin, Germany (2016).
13. Shen, Y., Huang, X.: Attention-Based Convolutional Neural Network for Semantic Relation Extraction. *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers, The COLING 2016 Organizing Committee*, pp. 2526–2536, Osaka, Japan (2016).
14. Schuster, S., Manning, C., D.: Enhanced English universal dependencies: an improved representation for natural language understanding tasks. In: *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16), ELRA*, pp. 2371–2378, Portorož, Slovenia(2016).
15. Kiavash, S., Rigel, G., Venkatakrishnan, V., N.: Extractor: Extracting attack behavior from threat reports. In: *IEEE EuroS&P*, pp. 598–615, IEEE, Vienna, Austria, (2021), <https://doi.org/10.1109/EuroSP51992.2021.00046>
16. Gao, P., Shao, F., Liu, X., Xiao, X., Qin, Z., Xu, F., Mittal, P., Kulkarni, S., Song, D.: Enabling efficient cyber threat hunting with cyber threat intelligence. In: *2021 IEEE 37th International Conference on Data Engineering (ICDE)*, pp. 193–204, IEEE, San Sebastian (2021).

17. Gasmi, H., Laval, J., Bouras, A.: Information extraction of cybersecurity concepts: an lstm approach. *Applied Sciences*, vol. 9, (2019).
18. Zhao, J., Yan, Q., Liu, X., Li, B., Zuo, G.: Cyber threat intelligence modeling based on heterogeneous graph convolutional network. In: *In Proceedings of the 23rd international symposium on research in attacks, intrusions and defenses (RAID 2020)*, pp. 241–256, USENIX, San Sebastian (2020).
19. Pingle, A., Piplai, A., Mittal, S., Joshi, A., Holt, J., Zak, R.: RelExt: relation extraction using deep learning approaches for cybersecurity knowledge graph improvement. *ASONAM '19: Proceedings of the 2019 IEEE/ACM International Conference on Advances in Social Networks Analysis and Mining*, vol. 2, pp. 879–886, ACM, Vancouver, British Columbia, Canada (2020).
20. Jo, H., Lee, Y., Shin, S.: Vulcan: Automatic extraction and analysis of cyber threat intelligence from unstructured text. *Computers & Security*, vol. 120, (2022).
21. Marchiori, F., Conti, M., Verde, V.: STIXnet: A Novel and Modular Solution for Extracting All STIX Objects in CTI Reports. *ARES 23: Proceedings of the 18th International Conference on Availability, Reliability and Security*, vol. 2, pp. 1–11, ACM, Benevento, Italy (2023).
22. Li, Y., Guo, Y., Fang, C., Liu, Y., Chen, Q.: A Novel Threat Intelligence Information Extraction System Combining Multiple Models. *Security and Communication Networks*, (2022), <https://doi.org/10.1155/2022/8477260>
23. Sarhan, I., Spruit, M.: Open-CyKG: An Open Cyber Threat Intelligence Knowledge Graph. *Knowledge-Based Systems*, vol. 233, (2021), <https://doi.org/10.1016/j.knosys.2021.107524>
24. Piplai, A., Mittal, S., Joshi, A., Finin, T., Holt, J., Zak, R.: Creating Cybersecurity Knowledge Graphs From Malware After Action Reports. in *IEEE Access*, vol. 8, pp. 211691–211703, (2020), <https://doi.org/10.1109/ACCESS.2020.3039234>
25. Wang, G., Liu, P., Huang, J., Bin, H., Wang, X., Zhu, H.: KnowCTI: Knowledge-based cyber threat intelligence entity and relation extraction. *Computers & Security*, vol. 141, (2024).
26. You, Y., Jiang, Z., Zhang, K., Feng, H., Jiang, J., Yang, P.: TiGNet: Joint entity and relation triplets extraction for APT campaign threat intelligence. *27th International Conference on Computer Supported Cooperative Work in Design (CSCWD)*, pp. 1687–1694. IEEE, Tianjin, China (2024), <https://doi.org/10.1109/CSCWD61410.2024.10580395>
27. Liu, Y., Han, X., Zuo, W., Lv, H., Guo, J.: CTI-JE: A Joint Extraction Framework of Entities and Relations in Unstructured Cyber Threat Intelligence. *27th International Conference on Computer Supported Cooperative Work in Design (CSCWD)*, pp. 2728–2733. IEEE, Tianjin, China (2024), <https://doi.org/10.1109/CSCWD61410.2024.10580210>
28. Mouiche, I., Saad, S.: TI-NERmerger: Semi-automated Framework for Integrating NER Datasets in Cybersecurity. In *Proceedings of the 21st International Conference on Security and Cryptography*, vol. 1, pp. 357–370, SciTePress, Dijon, France (2024)
29. Mouiche, I., Merboub, H., Saad, S.: Context-aware Entity-Relation Extraction Pipeline for Threat Intelligence Knowledge Graphs. *TechRxiv*, (2025) <https://doi.org/10.36227/techrxiv.173627493.39916970/v1>
30. Wang, X. et al.: DNRTI: A Large-Scale Dataset for Named Entity Recognition in Threat Intelligence. *2020 IEEE 19th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, pp. 1842–1848. IEEE, Guangzhou, China, (2020), <https://doi.org/10.1109/TrustCom50675.2020.00252>